

Literature Benchmark Brief: Kim_2024_ApplEnergy

Title: Accuracy of a short-term wind power forecasting model based on deep learning using LiDAR-SCADA integration

Authors: D. Kim, G. Ryu, C. Moon (2024)

Venue: Applied Energy | **DOI:** 10.1016/j.apenergy.2024.123882

Cluster: Cluster A: Spatio-Temporal Wind/Wake

Core Theoretical Mechanism & Model Formulation

Deep neural fusion of look-ahead nacelle LiDAR and multi-channel 10-min SCADA

Empirical Scale & Experimental Dataset

400-MW Anholt offshore wind farm (111 Siemens 3.6-MW turbines)

Relevance to IEEE TSTE & Sustainable Energy Systems

Demonstrates utility of upstream aerodynamic preview in mitigating forecast error

Identified Gaps & Operational Vulnerabilities

Does not address SCADA packet loss or latency degradation

Role in Manuscript Defense & Reviewer #2 Proof

Supports our dynamic wake graph as an algorithmic surrogate for missing spatial LiDAR